LOCAL AREA NETWORK (LAN) SETUP AND CONFIGURATION

Project Description

This project demonstrates the setup and configuration of a Local Area Network (LAN) using Cisco Packet Tracer. The network connects multiple devices such as PCs, a switch, a printer, and a server to simulate a small office or home network environment. Each device is assigned an IP address to enable communication within the network. The project also includes testing connectivity using the ping command and accessing network services such as an FTP server. This helps in understanding basic networking concepts like LAN topology, IP addressing, and device communication within a local network.

Objectives

- To understand the concept of a Local Area Network (LAN).
- To create a network using Cisco Packet Tracer.
- To connect multiple devices through a switch.
- To configure IP addresses for network devices.
- To test connectivity between devices using the ping command.
- To access services such as FTP within the network.

Tools Used

Cisco Packet Tracer

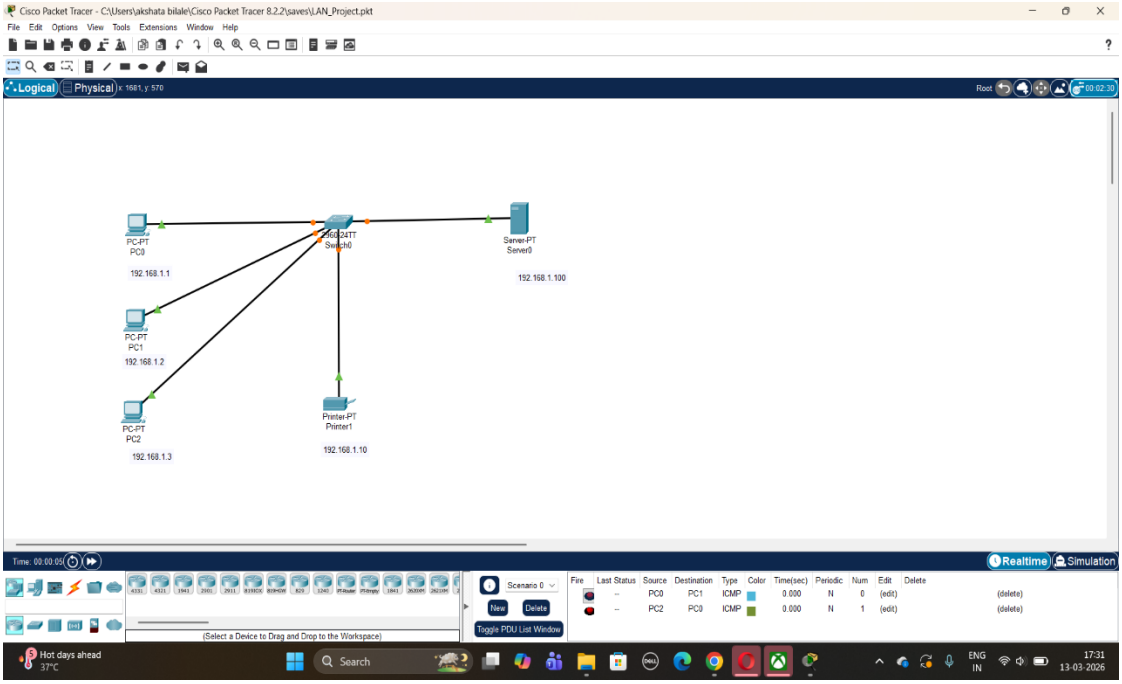
A network simulation software used to design, configure, and test the Local Area Network (LAN) setup virtually.

Network Topology

The network in this project was created using Cisco Packet Tracer and follows a **star topology**, where all devices are connected to a central switch.

Devices used in the network:

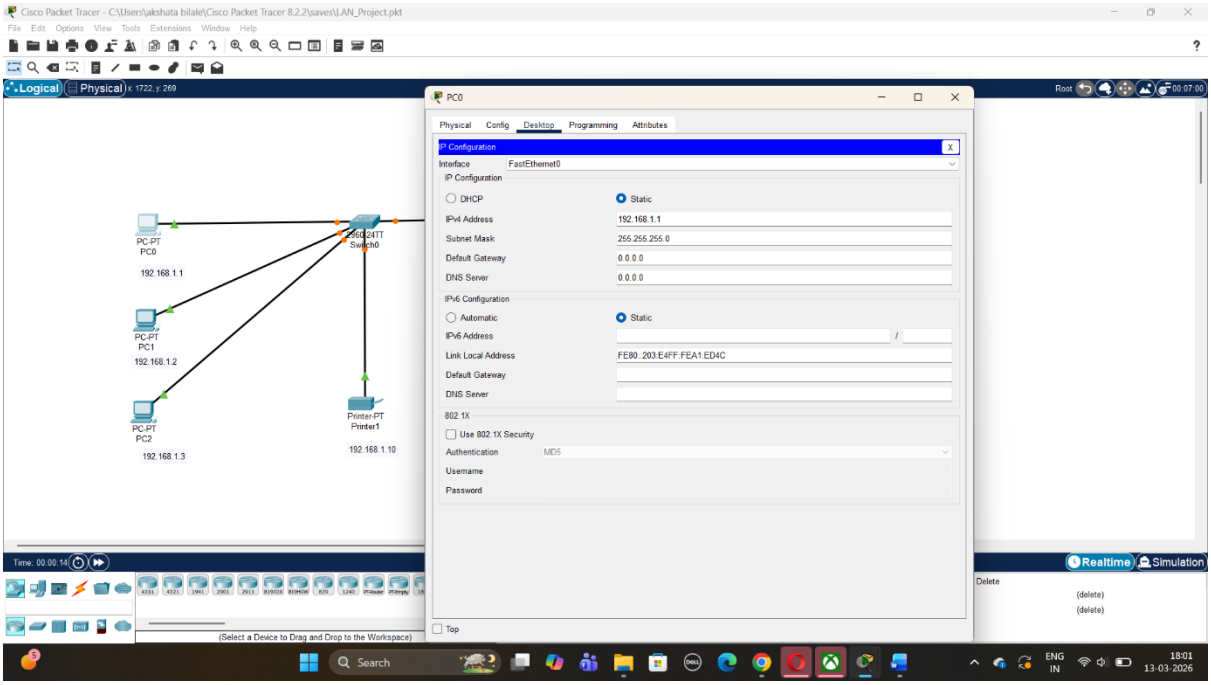
- **3 PCs** – PC0 (192.168.1.1), PC1 (192.168.1.2), PC2 (192.168.1.3)
- **1 Switch** – Cisco 2960 Switch used as the central connecting device
- **1 Printer** – Printer1 (192.168.1.10) for network printing
- **1 Server** – Server0 (192.168.1.100) used for services like file sharing (FTP)

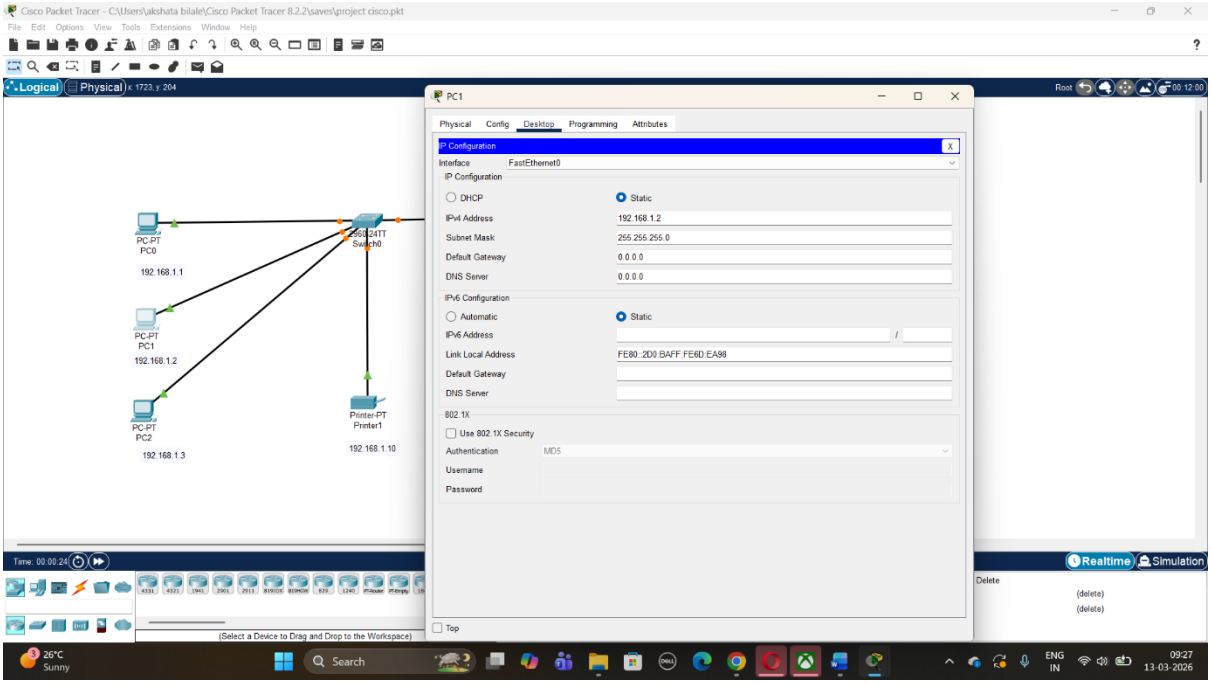


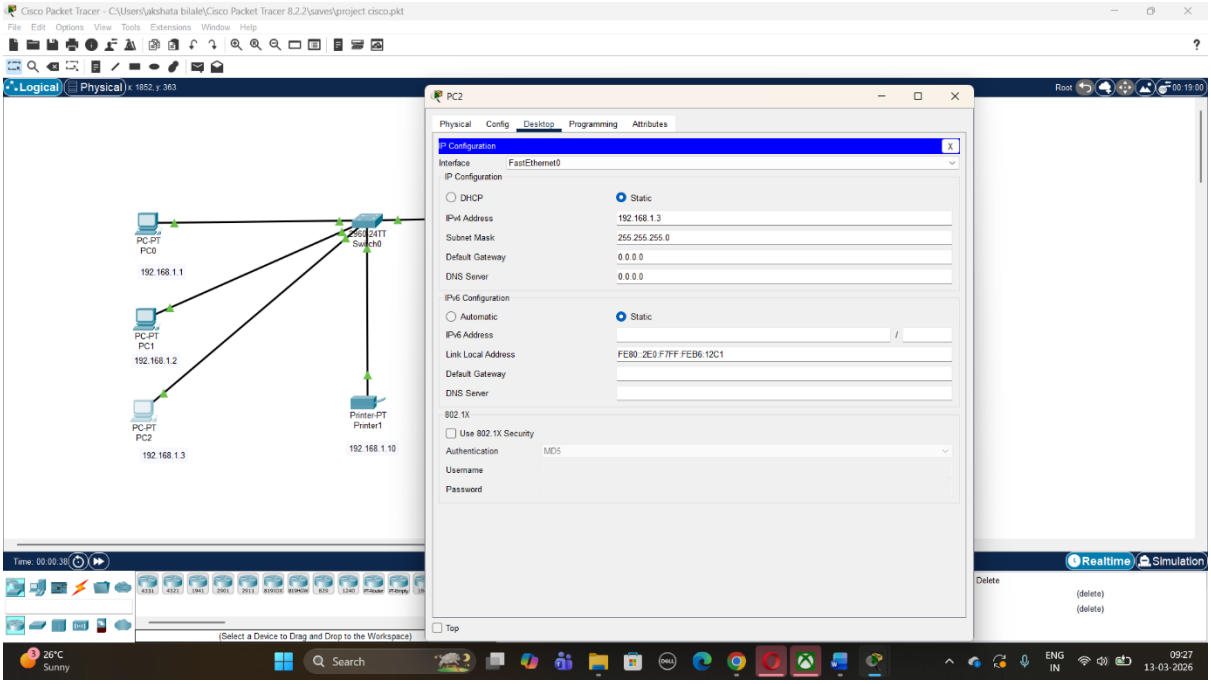
IP Configuration of All PCs

Each PC in the network was configured with a unique IP address within the same network range **192.168.1.0/24**. This configuration allows all devices to communicate within the Local Area Network created using Cisco Packet Tracer.

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.1.1	255.255.255.0	Not Required
PC1	192.168.1.2	255.255.255.0	Not Required
PC2	192.168.1.3	255.255.255.0	Not Required





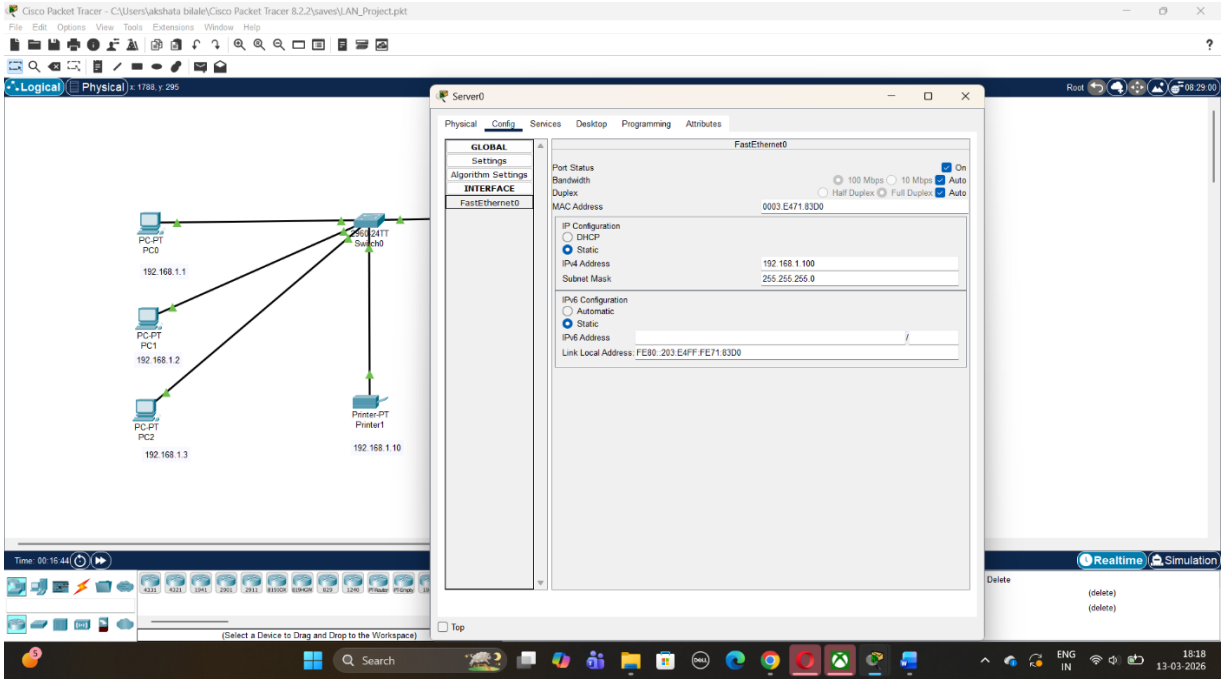


IP Configuration of Server and Printer

Server0

IP Address: 192.168.1.100

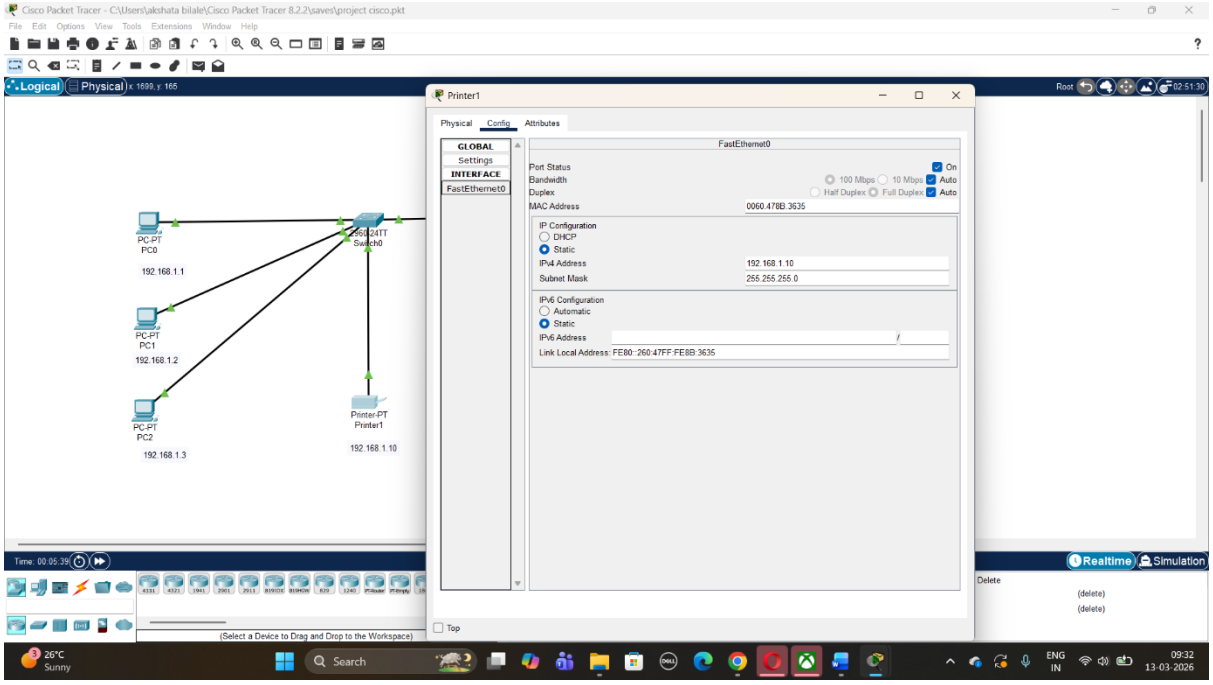
Subnet Mask: 255.255.255.0



Printer1

IP Address: 192.168.1.10

Subnet Mask: 255.255.255.0



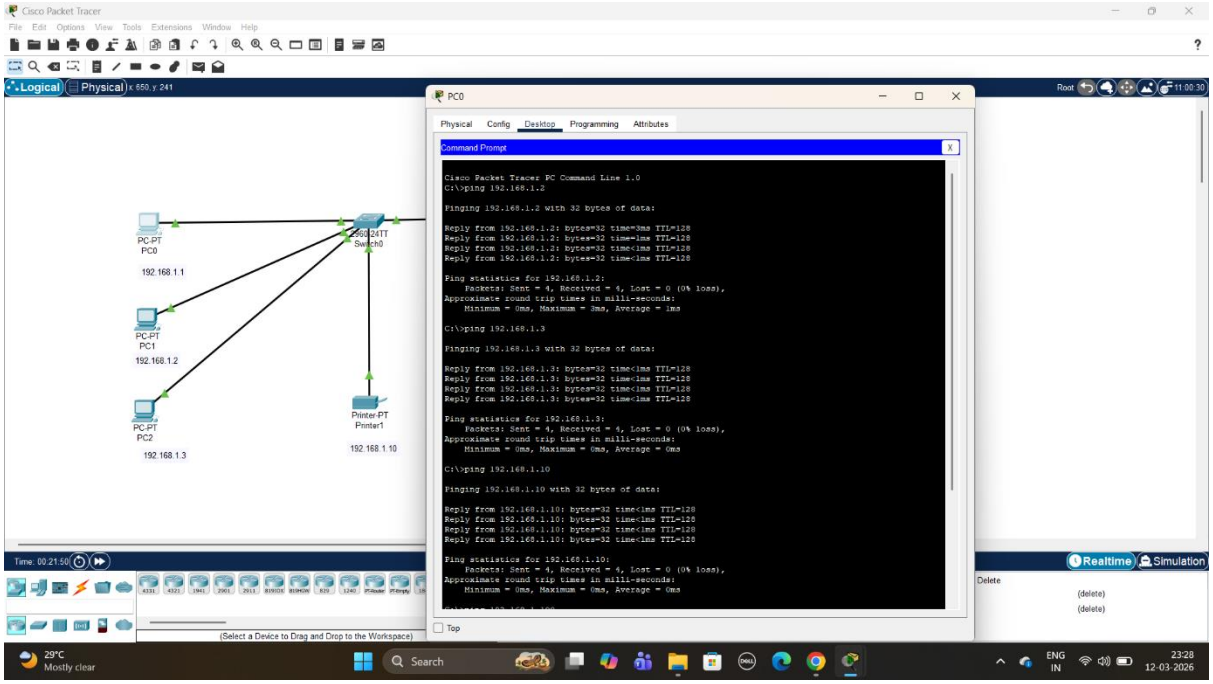
Network Testing

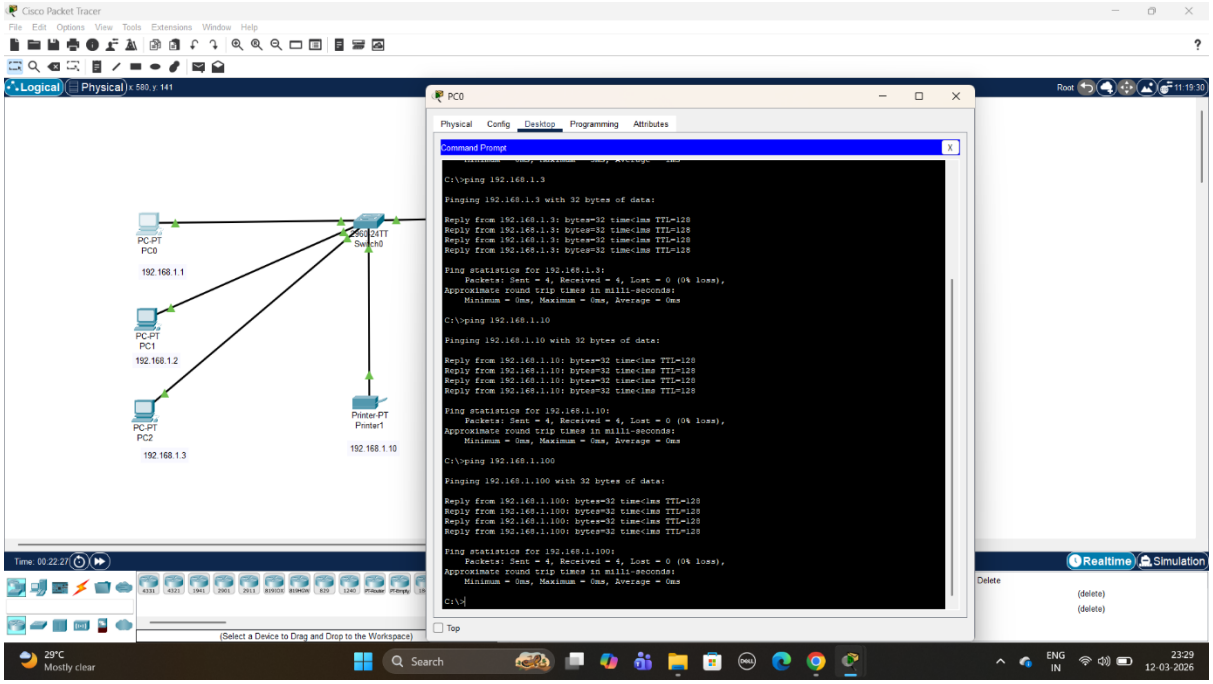
After configuring the devices, connectivity was tested using the ping command from the command prompt.

ping 192.168.1.2

ping 192.168.1.3

Successful replies confirmed that devices were able to communicate with each other within the network.





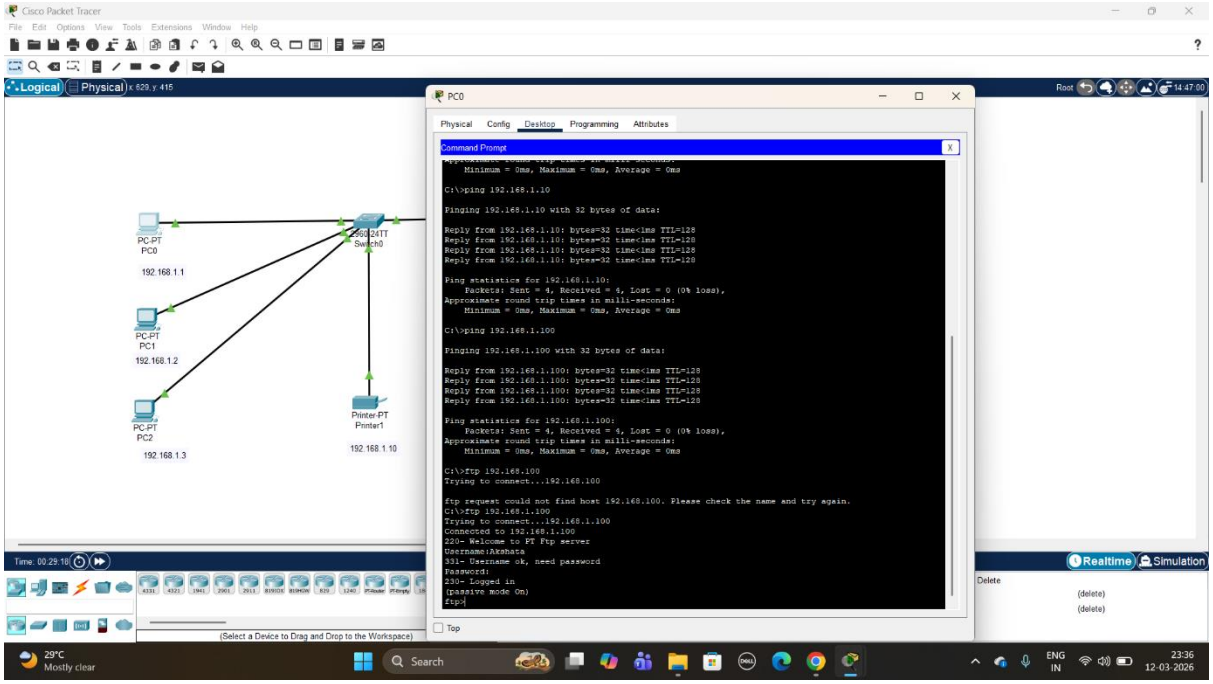
FTP Server Access

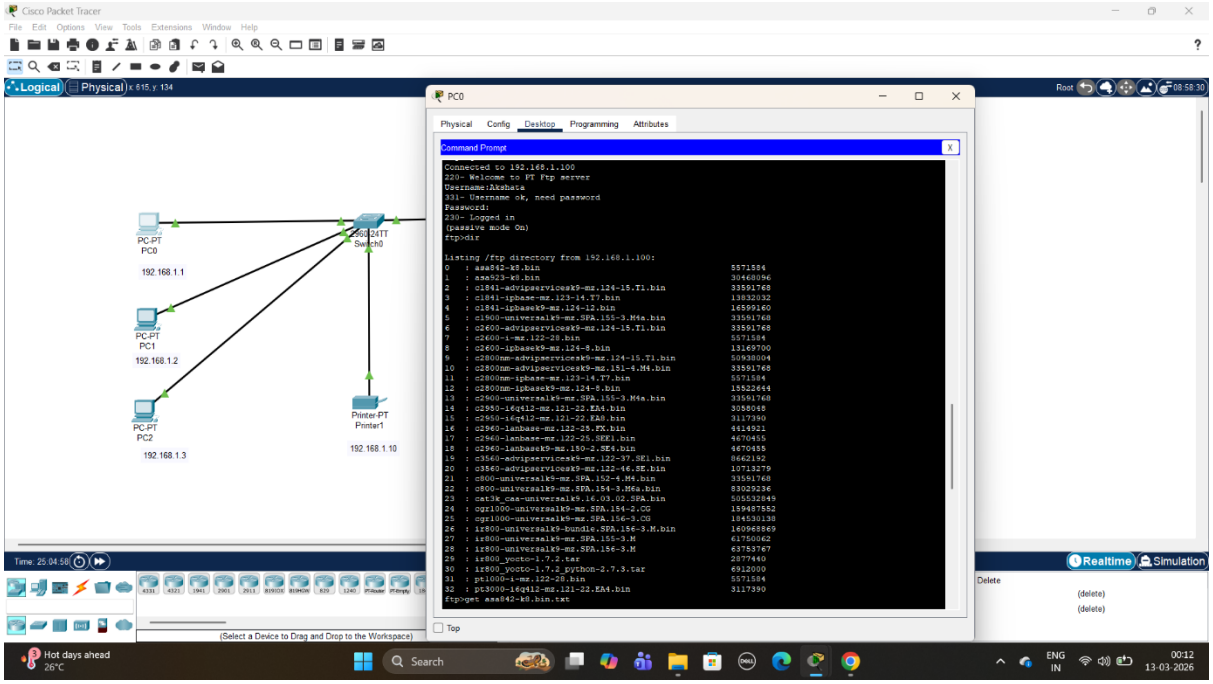
An FTP server was configured in the network to demonstrate file transfer functionality.

Steps performed:

1. Opened the command prompt on a PC.
2. Used the FTP command to connect to the server.
3. Logged in using server credentials.
4. Displayed the directory contents using the **dir** command.

This confirmed that the FTP service was working properly in the LAN.





Results

The LAN network was successfully created using Cisco Packet Tracer. All devices were connected properly through the switch and assigned appropriate IP addresses. Communication between devices was verified through ping tests, and the FTP server was accessed successfully.

This demonstrated proper functioning of the local network.

Conclusion

This project helped in understanding the setup and configuration of a Local Area Network using Cisco Packet Tracer. It demonstrated how multiple devices can be connected through a switch and communicate with each other using IP addressing. The project also provided practical experience with network configuration, testing connectivity, and accessing network services within a simulated environment.